

VOYAGER: An Open-Ended Embodied Agent with LLMs

Problem and Motivation

Training embodied agents that can operate in open-ended environments has long been a challenge. Traditional reinforcement learning agents are often constrained by sparse rewards, limited sample efficiency, and weak generalization to unseen tasks. On the other hand, large language models (LLMs) have shown strong zero-shot abilities in textual domains. This paper investigates whether LLMs can be adapted to the setting of *lifelong embodied learning* and *open-ended exploration*.

The authors focus on the **Minecraft** environment due to its rich, complex dynamics and wide task diversity. The goal is to design an agent that can continually explore, acquire new skills, reuse knowledge, and adapt to novel tasks. This complex and challenging task motivates the authors to utilize Large language models, particularly GPT-4, to solve the task due to their ability to generate structured code, reason over symbolic representations of the environment, and handle language-based instructions.

Related Works

Prior work like ReAct [1], Reflexion [2], and AutoGPT [3] integrates LLMs with action-execution loops for planning. However, these systems typically rely on short-term prompting and lack structured memory or skill abstraction. They often fail under noisy feedback and cannot scale to open-world environments.

Code-generation agents such as CodeX [4] offer stronger abstraction but lack the ability to iteratively refine or adapt programs through interaction. Most prior RL-based approaches to Minecraft require expert demonstrations, dense rewards, or task-specific engineering, which makes them unsuitable for general-purpose open-ended learning.

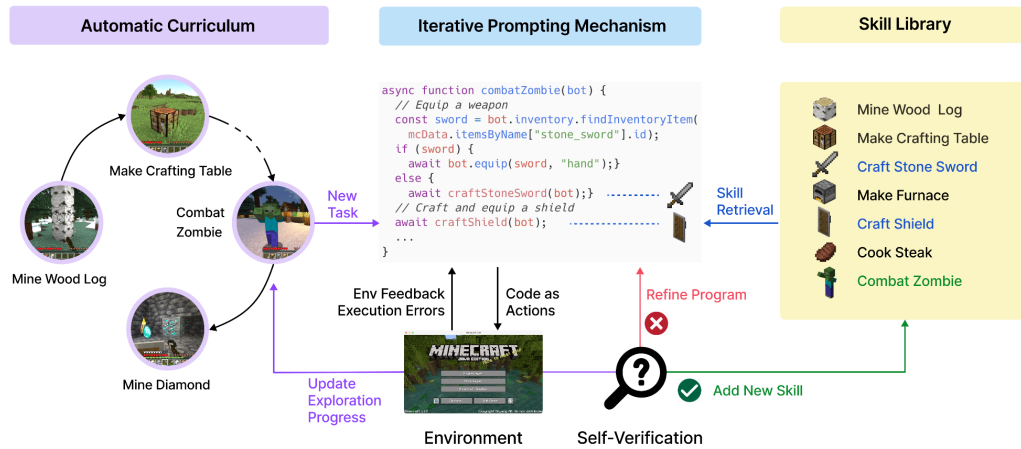
Solution Overview

The authors propose **VOYAGER**, a lifelong learning agent built around GPT-4 that interacts with the Minecraft environment through a novel "**program-as-action**" paradigm. The key components include:

- **Program-as-Action Interface:** Instead of primitive actions, the agent writes MineDojo [5]-compatible Python programs to perform complex tasks. These programs are interpretable, composable, and reusable.
- **Self-Adaptive Curriculum:** VOYAGER uses GPT-4 to propose new goals based on current knowledge, inventory, and environment state. This supports autonomous curriculum building without requiring predefined tasks or external rewards.

- **Iterative Code Refinement:** The system uses environment feedback (e.g., error traces, execution failures) and additional GPT-4 passes to refine the generated programs, improving robustness and enabling self-correction.
- **Skill Library with Retrieval:** Successfully validated programs are stored in a vector-indexed skill library using text embeddings (with OpenAI's text embedding model). The agent retrieves and reuses these skills in new contexts, supporting generalization and memory.

These components enable the agent to discover new items, unlock tech trees, and explore complex worlds significantly faster than baselines.



Limitations

While VOYAGER shows impressive performance, it has several limitations:

- **Dependency on LLM-Generated code:** This introduces high computational cost and potential brittleness in cases of hallucinated code generation.
- **Symbolic World Representation:** The agent operates on text-based world states rather than raw visual input, limiting generality to environments with clean APIs.
- **Verification Limitations:** Task success is determined via heuristic checking and GPT-4-based evaluation, which may miss subtle errors or misinterpret game state.
- **Skill Library Scalability:** As the number of stored programs grows, retrieving and composing relevant skills becomes increasingly difficult, potentially introducing inefficiencies in both memory usage and planning.
- **Lack of policy optimization:** The system does not learn optimal action policies through reinforcement or gradient-based updates.

Future Directions

The authors propose several promising directions to extend this work:

- **Multimodal Integration:** Combining VOYAGER with vision-language models could enable raw visual grounding, allowing deployment in real-world or visually complex environments.
- **Hierarchical Skill Composition:** Developing mechanisms to plan over sequences of skills and build abstract task graphs would enable more advanced and modular reasoning.
- **Human Feedback Incorporation:** Incorporating interactive feedback or imitation from human users could guide exploration and improve robustness.
- **Cross-Environment Transfer:** Applying the VOYAGER framework to other open-ended simulators would test its generality in open-world agents.

Summary of Class Discussion

[Limitation] Text-based limitation in a visual environment: Although Minecraft is a visually rich, spatial world, VOYAGER interacts with it solely through structured API calls and symbolic inputs. It does not perceive or reason over raw visual inputs.

[Limitation] Lack of spatial awareness: The agent's internal representations lack true spatial grounding. This raised the broader question of what constitutes a meaningful representation of Minecraft for embodied agents operating in open-ended 3D environments.

[Future Direction] Spatial augmentation: Professor You suggested augmenting the system with navigation modules or spatial adaptors to give the agent a better sense of location and orientation. However, limitations related to context length and token efficiency in LLMs make it difficult to encode rich spatial states.

[Future Direction] Skill library as a potential improvement area: The class discussed extending the current definition of the skill library to include spatial information or more diverse, composable behaviors. This could help mitigate VOYAGER's limitations without requiring full multimodal finetuning.

[Limitation] No direct learning from videos: VOYAGER does not directly learn from video demonstrations. Videos are used only for visualization purposes, limiting the agent's ability to imitate human-like behavior from visual data.

[Future Direction] Opportunity for synthetic trajectory data: Unlike real-world robotics, collecting synthetic demonstration data in Minecraft is cheap and scalable. This opens up opportunities to train agents using large-scale, human-like trajectories generated in simulation.

[Limitation] General limitations of reinforcement learning: The class noted that traditional RL techniques often struggle in open-ended environments. LLM-based reasoning may offer more generalizable and sample-efficient alternatives in this context.

[Future Direction] Industry relevance and future potential: The class observed that similar approaches are already being pursued in industry. The convergence of generative models, virtual agents, and open-ended game environments is attracting significant investment and may reshape how intelligent agents are designed in games and simulations.

References

- [1] Yao, Shunyu, et al. "React: Synergizing reasoning and acting in language models." International Conference on Learning Representations (ICLR). 2023.
- [2] Shinn, Noah, et al. "Reflexion: Language agents with verbal reinforcement learning." Advances in Neural Information Processing Systems 36 (2023): 8634-8652.
- [3] Richards, Toran Bruce. "Significant-gravitas/autogpt: An experimental open-source attempt to make gpt-4 fully autonomous., 2023." URL <https://github.com/Significant-Gravitas/AutoGPT>.
- [4] Chen, Mark, et al. "Evaluating large language models trained on code." arXiv preprint arXiv:2107.03374 (2021).
- [5] Fan, Linxi, et al. "Minedojo: Building open-ended embodied agents with internet-scale knowledge." Advances in Neural Information Processing Systems 35 (2022): 18343-18362.

MINEDOJO: Building Open-Ended Embodied Agents with Internet-Scale Knowledge

Problem and Motivation

Compared with the last paper, this paper extends beyond the text-only scenarios and uses a video language model as a backbone for playing gr

While specialist RL agents excel at tasks such as Atari, Go, and robotics, they falter in **open-ended domains** with vast complexity and little reward shaping. Human learning, by contrast, thrives on open exploration and external knowledge. **MINEDOJO** sets out to build **generalist embodied agents** that leverage internet-scale data within a highly expressive world: Minecraft. Specifically, the authors highlight the need for:

1. A simulator that supports a near-infinite range of goals.
2. A massive, multimodal knowledge base for scalable learning.
3. A flexible, transformer-based agent architecture to incorporate large-scale data effectively.

Compared with the last paper, MINEODJO leverages multimodal backbones and data, rather than relying on textual data exclusively.

Related Works

Earlier Minecraft benchmarks (e.g., MineRL [1], BASALT [2]) focus on narrower tasks with curated demonstrations or dense rewards, limiting scalability. Although large (video-)language models (e.g., CLIP [3], SayCan [4], AlphaStar [5]) can ground text in action, they typically only handle simpler tasks or smaller domains

MineDojo advances this line of work by combining internet-scale knowledge (YouTube, Wiki, Reddit) with a standardized simulation platform and a learned reward model that enables open-vocabulary, multi-task reinforcement learning.

Solution Overview

MineDojo is a comprehensive framework for open-ended embodied agent learning, built on Minecraft. It has three main contributions:

1. Simulation Platform with Diverse Tasks

- Over 1,500 **Programmatic Tasks** ranging from survival challenges to harvesting, tech-tree progression, and combat. These have automated success criteria.
- Over 1,500 **Creative Tasks** with open-ended goals (e.g., “build a haunted house”), using language prompts. Because they have no straightforward success condition, the authors propose a learned metric for evaluation.

2. Internet-Scale Knowledge Base

- Comprises 730K+ Minecraft YouTube videos, 6.7K Wiki pages, and 340K+ Reddit posts.
- Enables agents to leverage the collective wisdom of the gaming community instead of relying on expert demonstrations or dense reward shaping.

3. MINECLIP: Learned Reward Model

- A contrastive video-language model that aligns free-form text instructions with corresponding video segments.
- Produces a continuous reward signal for open-ended tasks based on how well the agent's behavior matches the textual goal.
- Significantly correlates with human judgments in creative tasks.

This framework enables an agent to learn across many tasks, generalize to unseen contexts, and adapt to new goals with minimal manual reward shaping or curated demonstrations.



Limitations

Despite its breadth and potential, **MINEDOJO** faces several challenges:

1. **Lack of Spatial Awareness**

As the class discussed, it is hard to train a multimodal backbone with good spatial awareness, and they merely learn some associations.

2. **Symbolic or Structured Representations**

Current tasks mostly rely on text-based feedback rather than full raw-pixel observations, which can impede transfer to real-world domains.

3. **Approximate Evaluation for Creative Tasks**

Success in free-form goals is estimated via MINECLIP's learned reward, which may not perfectly capture nuanced objectives.

4. **Computational Cost**

Training with large models (e.g., MINECLIP) can be resource-intensive

Future Directions

The paper outlines several avenues for future research:

- **Multimodal grounding** with vision-language models to operate on raw pixel inputs, enabling application to real-world settings.
- **Open-source alternatives** to the old backbone used in the paper (e.g., more recent ones like LLaMA, Mistral) to improve accessibility and reproducibility.
- **Hierarchical planning and memory** to support long-horizon tasks and compositional skill reuse.
- **Human feedback integration**, allowing the agent to learn interactively and refine behaviors through social cues.

Class Discussion

[Mechanism Focus] MineCLIP's Contrastive Video-Language Reward

The discussion centered on MineCLIP, a model that learns to map YouTube videos and transcripts onto free-form text goals, enabling open-vocabulary, dense reward signals. Students and Professor You emphasized that the alignment across modalities—video plus language—is crucial and leverages structured YouTube content (in contrast to single-modal data from Wiki or Reddit).

[Choice of Minecraft] Flexible and Standardized Action Space

While Minecraft's low-resolution visuals may ease video processing, its real strength lies in a

vast action space exposed through the MineDojo API. This standardized, text-based interface simplifies both reinforcement learning and goal specification.

[Clarification] Primary Use of Video Data

It was highlighted that MineCLIP primarily uses video data for training supervision of the learned reward model rather than textual data only (like the last paper).

[Future Direction] Broader Trends in Multimodal Learning

The class noted that MineDojo's approach mirrors a rising wave of multimodal research, where contrastive learning techniques integrate video and text. Extending these methods beyond Minecraft could enable more robust world modeling in other open-ended domains.

[Industry Relevance] Growing Investment in VLMs and Simulation

Professor You mentioned that the concept of using contrastive video-language models for dynamic world modeling is of keen interest to industry, illustrated by startups (e.g., "WorldAI") raising substantial funding (up to \$230M). This signals strong commercial momentum for agents blending LLM-based reasoning with virtual environments.

[1] Minerl: A large-scale dataset of minecraft demonstrations

[2] The minerl basalt competition on learning from human feedback.

[3] Learning transferable visual models from natural language supervision.

[4] Do as i can, not as i say: Grounding language in robotic affordances.

[5] Alphastar: Mastering the real-time strategy game starcraft ii